Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/03/2014
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Proteomics Analysis of Arabidopsis Seedlings in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Rapid Turn Around Flight		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Luesse, Darron Robert (Southern Illinois University, Edwardsville)		
Grant/Contract No.:	NNX13AM48G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The space flight environment provides a unique environment to understand how gravity informs plant growth, development, and physiological processes. The central objective of this study is to determine what proteins are differentially expressed during space flight in developing Arabidopsis seedlings. A body of research has evaluated the differential gene expression on space flight vs ground controls, but gene expression provides an indirect measure of what proteins are expressed. Genes may be transcribed but the transcripts degraded, or translated proteins may be regulated through post translational modifications. Our hypothesis is simple: Some components of plant physiology that are informed by gravity are regulated by post transcriptional or post translational mechanisms. Proteomic analysis is the best/only method to identify these components. We propose to use BRIC-PDFU hardware on a Rapid Turn-Around Space Flight Experiment to gain insights into differences in protein profiles between Arabidopsis seedlings grown		

	during space flight and ground based controls. Arabidopsis seedlings will be germinated, grown, and tissues fixed on orbit. Once returned to Earth, total protein will be extracted, labeled with iTRAQ reagents, and analyzed for differences in protein profiles. These data will complement the gene expression data currently available from space flight experiments and also provide novel insights by supplying data on differences in post-transcriptional regulation. They will also provide a foundation for a network analysis to generate additional hypotheses as to the mechanisms involved in gravity perception/response in plants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research may lead to the discovery of novel components that determine how plants respond to gravity and to the space flight environment. It will also form a foundation for the establishment of hypotheses to support research on plant improvement for space flight and life support in the space flight environment.
Task Progress:	We have made good progress toward our goals for space flight. We have completed the Definition phase of our project, have determined key parameters that will be necessary for the space flight opportunity, and have satisfied all requirements to move into the initial testing prior to space flight. We are currently in the process of finalizing preparations for the Science Verification Test.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Wyatt SE, Shen K, Hayden M, Fenstemaker S, Tucker A, Moore M, Brita A. "Plant gravitropic signal transduction: A network analysis leads to gene discovery." Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. , Aug-2014
Abstracts for Journals and Proceedings	Wyatt SE, Luesse DR. "Proteomics Analysis of Arabidopsis Seedlings in Microgravity." 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. , Jun-2014
Abstracts for Journals and Proceedings	Basu P, Hayden M, Wyatt SE. "Preparing for a spaceflight: Definition Phase." Midwestern Section of the American Society of Plant Biologists, Columbus, OH, March 22-23, 2014. Midwestern Section of the American Society of Plant Biologists, Columbus, OH, March 22-23, 2014. , Mar-2014
Books/Book Chapters	Basu P, Luesse DR, Wyatt SE. "Proteomic approaches and their application to plant gravitropism." in "Methods in Molecular Biology." Ed. E.B. Blancaflor. In press as of July 2014., Jul-2014